Machine Learning Explainability for Intrusion Detection in the Industrial Internet of Things

Love Allen Chijioke Ahakonye, Cosmas Ifeanyi Nwakanma, Jae Min Lee, and Dong-Seong Kim

Abstract

Intrusion and attacks have consistently challenged the Industrial Internet of Things (IIoT). Although artificial intelligence (AI) rapidly develops in attack detection and mitigation, building convincing trust is difficult due to its black-box nature. Its unexplained outcome inhibits informed and adequate decision-making of the experts and stakeholders. Explainable AI (XAI) has emerged to help with this problem. However, the ease of comprehensibility of XAI interpretation remains an issue due to the complexity and reliance on statistical theories. This study integrates agnostic post-hoc LIME and SHAP explainability approaches on intrusion detection systems built using representative AI models to explain the model's decisions and provide more insights into interpretability. The decision and confidence impact ratios assessed the significance of features and model dependencies, enhancing cybersecurity experts' trust and informed decisions. The experimental findings highlight the importance of these explainability techniques for understanding and mitigating IIoT intrusions with recourse to significant data features and model decisions.

Introduction

The Internet of Things (IoT) has become an increasingly popular trend poised for prospective transformation and revolution of routine tasks by utilizing billions of intelligent devices estimated to reach around 75 billion by 2025 [1]. Although IoT attacks are rising, technology is becoming increasingly important in manufacturing, transportation, agriculture, and healthcare. However, rapid IoT deployment in industries has led to numerous vulnerabilities. An example is the attack that led to many Internet services, like Twitter and Amazon, being unavailable for several hours [1], causing extensive financial loss and colossal damage. A recent report states that the financial damage from IoT attacks will amount to about US20 billio by 2021 [1]. Similarly, IoT analytics, in 2022, had no major security incidents, thanks to recent security research and solutions investments. However, critical vulnerabilities still prevailed, leading to the issuing of corresponding advisories.

As IIoT technology and critical infrastructure expand and integrate, Artificial Intelligence (AI) solutions based on attack intelligence become crucial for cybersecurity defence [2]. There are various measures in IIoT network security, but their effectiveness could be improved, particularly in light of the rising threat landscape. Machine Learning (ML) and Deep Learning (DL) techniques uncover covert attack patterns, often beyond human security professionals' reach. It highlights the need for research that explains cybersecurity attacks, emphasizing the importance of understanding the reliability of these models' decisions in critical settings [1]. It can be challenging to trust the predictions of such opaque models.

ML-based IDSs should provide clear and trustworthy explanations for their predictions. It ensures user confidence and understanding, even for individuals without advanced mathematical or AI knowledge [3]. Explainable AI (XAI) aims to improve transparency in AI models' predictions, addressing the need for comprehensibility and trustworthiness, especially in complex domains like cybersecurity [4, 5]. With significant progress in introducing transparency and explainability, challenges related to ease of comprehension and trust still exist in XAI research.

Post-hoc explainability techniques aim to enhance understanding and trust in complex black-box ML models, making them suitable for real-world deployment. This study focuses on the SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) [5]. Despite their wide usage and support in the XAI community, model agnosticism and visualization tools effectively communicate the model's behavior. It focuses on local interpretability and a global perspective on feature importance with a solid mathematical foundation that measures the contribution of each feature. These qualities provide a quantitative and understandable explanation, addressing trust and reliability concerns. Regardless of type or structure, their applicability to any ML model improves interpretability and trust among users, security experts, and stakeholders, even without extensive mathematical or AI expertise.

Specifically, the contributions of this study are as follows:

1. Analysis of the performance of state-of-the-art algorithms for attack detection based on the f-measure evaluation metrics due to imbalanced attack class samples.

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2. Post-hoc technique for explainability insight following model prediction based on the essential justifications of feature inputs to the model predictions.
3. The explainability of two (2) post-hoc XAI approaches, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Ex-planations (LIME) evaluated state-of-the-art predictive algorithms.
4. An experimentation of quantitative metrics of the decision and confidence impact ratios to evaluate the trust and concordance of the LIME and SHAP explainability techniques on the prediction algorithms.
5. A comparative analysis of the SHAP and LIME interpretation of the prediction models regarding decision and confidence impact ratios, mean vote for model inference, trust, and enhanced decision-making.

The organization of the study is thus: we review existing related work. We discuss the specification and development of the proposed XAI architecture. We discuss the experimentation of the XAI architecture and results. We conclude the study.

Related Works

This section introduces recent research on the explainability of ML-based intrusion detection systems, emphasizing the essential contributions of prior studies. Notably, these studies underscore the crucial need for explainable post-hoc techniques in AI models across vital domains like transportation, healthcare, and industrial control systems [3–6]. Alhawawreh *et al.* [6] employed post-hoc XAI to unveil cyber-physical attack patterns, demystifying black box algorithms and offering practical insights for countering cyber threats. In the context of IoT attacks, [1] proposed an XAI system for deep learning-based IDSs. Despite being limited by the incomprehensibility of the model explainability to a non-expert user, the approach needed to solve the explainability challenge of model predictions. A model-agnostic explainer, TRUST XAI, relied on a statistical premise for transparency in IIoT [3]. While significant in a cybersecurity case study, it contradicted XAI's principle of comprehensibility, particularly for non-technical users [15].

A comparative analysis of three XAI approaches, LIME, RuleFit, and SHAP, highlighted their importance in building understanding and trust in deep learning solutions for IoT IDS [15]. However, it also brought the debate of "which of these three offers the best explanation for IIoT?" LIME and SHAP visualization offered the best explanation, as evident in their work and emphasized in our work in the subsequent sections. A two-stage pipeline involving XGBoost and an autoencoder trained on SHAP explanations to represent the model behavior during the training process [9]. Even with contributing to a level of explainability, the outcome needed to have the transparency expectation of XAI.

This study enhances feature relevance and local explanations in XAI for IIoT IDS. The post-hoc explainability approach employs agnostic SHAP and LIME techniques and a qualitative experimentation analysis to address the limitations of comprehensibility. The goal is to assess trust and concordance levels in IDS models for effective attack mitigation [11, 12]. By tackling the opacity of black-box ML models, this study elucidates the internal workings of these models, facilitating human understanding and regulatory control. Furthermore, the study critically examines the validity of these claims, evaluating whether insights derived from the XAI methods post-model deployment can genuinely enhance interpretability in IDS models and foster user trust.

Challenges persist despite significant strides in achieving explainability and transparency across various sectors. The explanations generated by XAI can be complex and underutilized, particularly in IoT IDS. The complexity of predictive models increases due to the evolving nature of cyber threats targeting IoT systems and the diverse data generated by the multitude of sensors and actuators. Table 1 summarizes some studies, highlighting the limitations and underscoring the improvements this research offers. The existing works show advancements in XAI comprehensibility with potential trust for complex IDS models.

IIoT Intrusion Detection XAI

Considering system architecture, data collection, and security, ML techniques are used for IIoT attack detection. Monitoring and updates are crucial for adapting to evolving attacks to ensure accurate detection and protection of IIoT critical infrastructures. XAI explainability is essential to enable trust and usability for decision-making [5] in IIoT applications offering insights into the decision-making process of complex intrusion detection models. These XAI methods improve model comprehensibility by providing quantitative interpretability, enabling stakeholders to understand the factors driving predictions in IIoT systems.

The schematic in Fig. 1 illustrates the procedural flow of the system model. In this study, intrusion detection and XAI techniques are crucial components of this architecture. The IDS takes inputs from IIoT sensors, actuators, output values, and learnable parameters, employing algorithms like Decision Trees, Random Forests, Transformers, and Deep Neural Networks. It analyzes the data to classify attacks and post-hoc LIME, and SHAP automatically extracts interpretation. LIME provides local insights for individual instances, and SHAP offers a global perspective across multiple cases. These techniques collectively distil essential input features the IDS uses to determine attack types. Quantitative evaluation of the LIME and SHAP interpretation for trustworthiness in the predictive model decisions indicates a dependence on the data features. The decision impact ratio gives the overall impact of the IIoT data features on the model's decision. In contrast, the confidence impact is the level of dependence on the prediction model.

XAI Model Explainability

For cybersecurity experts, AI models for intrusion detection should be understandable through XAI [4, 5] and enable them to believe in and modify such models. XAI intrusion detection model explainability approaches commonly used are from the Post-hoc and Ad-hoc categories [7]. The apparent design of algorithms in the ad-hoc approach enables transparency to provide clear and intuitive explanations from the ground up, such as decision trees, rule-based systems, rule extraction, and prototype-based interpretation for decision-making. Post-hoc explainability applies interpretability tech-

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Author	Year	IDS	XAI	IIoT Domain	ML	DL	Post-hoc SHAP	Post-hoc LIME	Decision Impact Ratio	Confidence Impact Ratio	Visualization	Ease of Comprehension
[1]	2022	√	√	√	χ	√	√	χ	χ	χ	√	√
[2]	2023	√	χ	√	√	√	χ	χ	χ	χ	χ	χ
[3]	2021	√	√	√	√	√	χ	χ	χ	χ	√	χ
[4]	2022	√	√	√	χ	√	χ	χ	χ	χ	χ	√
[5]	2023	√	√	√	√	√	√	√	χ	χ	√	√
[6]	2024	√	√	√	√	√	χ	√	χ	χ	√	√
[7]	2022	√	√	√	√	√	√	√	χ	χ	χ	χ
[8]	2022	√	√	√	√	√	√	√	χ	χ	√	√
[9]	2022	√	√	√	√	χ	√	χ	χ	χ	√	χ
[10]	2022	√	√	√	√	√	√	√	√	√	√	√
[11]	2023	√	√	√	√	χ	√	χ	χ	χ	√	√
[12]	2023	χ	√	√	χ	√	√	χ	χ	χ	χ	χ
[13]	2019	χ	√	√	√	χ	χ	χ	χ	χ	√	√
[14]	2019	√	√	√	χ	χ	χ	χ	χ	χ	χ	χ
This Study	2024	√	√	√	√	√	√	√	χ	√	√	√

TABLE 1. Summary of review of literature highlighting the key findings of each paper (Yes: √, No: χ).

niques after training a complex black-box model to provide insights into its intricacies and explain its predictions. This approach applies to any trained model without modifying the underlying architecture; hence, model-agnostic. Regarding essential explanations, such as feature contributions to the model output, post-hoc methods provide interpretation following model prediction.

Functionality of the Post-Hoc LIME

LIME takes in IIoT data points and employs a black-box attack attribution module to create a likelihood distribution for various attack types. During this process, feedback is temporarily halted while it calculates a weighted sum of the data points, revealing attack patterns. The assault attribution module leverages the techniques used by LIME to process this interruption and determine if it enhances the primary target class. This method employs a simple, linear, interpretable model to extract critical information based on probability and interruption data points. Subsequently, this extracted intelligence is organized and predicted as an attack.

Strengths: LIME is compatible with any algorithm and accommodates various data types, enhancing its applicability. This feature makes it suitable for various applications. It provides insights at a local level and explanations in an easily understandable format. LIME is a widely recognized and extensively used tool in XAI with support from multiple programming language libraries.

Limitations: LIME relies on generating perturbed data samples, which can result in varying quality of explanations based on sampling methods, potentially leading to unreliability.

Functionality of the Post-Hoc SHAP

The SHAP explainability method offers valuable insights into feature influence on machine learning models. It calculates contribution scores using Shapley values, providing global and local explanations. Global SHAP values reveal feature importance across the dataset, while local SHAP values explain a feature's impact on a single prediction. SHAP ensures consistency by attributing the total prediction value to feature contributions. It is model-agnostic, making it suitable for many ML models, including complex ones, and generates visualizations for interpreting feature contributions. This interpretability is crucial in applications requiring model transparency. Additionally, SHAP considers feature interactions, providing a holistic view of feature importance and interactions in predictions.

Strengths: It explains individual predictions without insight into the ML model machinery. Expressive solid power for the single predictions. Low insight into model internals. A high portability method that does not rely on the ML model specs to explain the predictions. Medium algorithmic complexity, easy to implement, it is challenging to get a complete understanding of the underlying concepts. It has a good level of comprehensibility and human-understandable explanations.

Limitation: The computational complexity of SHAP, mainly when dealing with high-dimensional data or complex models, is a significant limitation when applying this explainability technique. Hence, the LIME approach makes up for this challenge.

Explanation of the Post-Hoc LIME and SHAP Algorithm

In the LIME algorithm, perturbing the data x times creates modified instances with slight value changes around the original observation. LIME then uses this perturbed data to construct a local linear model. The perturbed data make predictions, and a similarity score is computed for each perturbed data point based on its distance from the original observation. From this perturbed data,

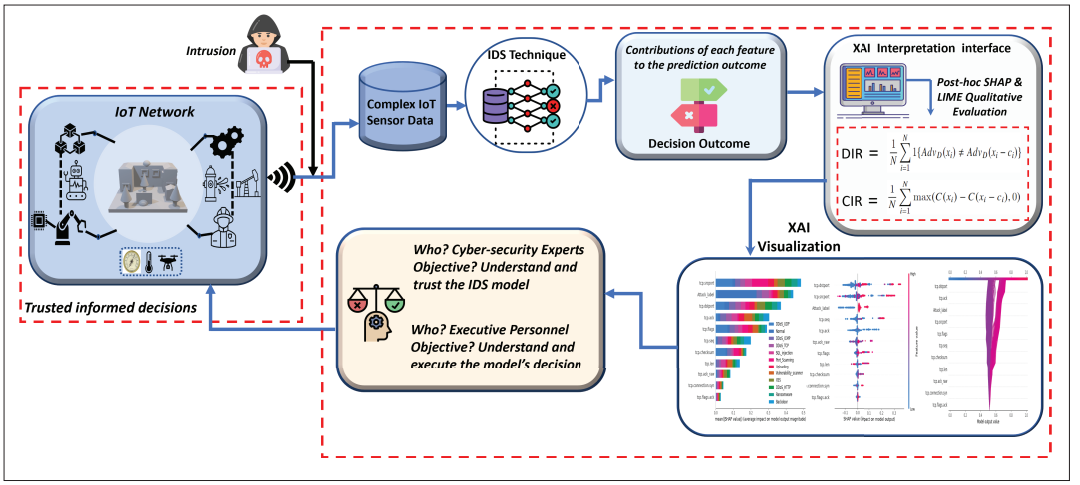


FIGURE 1. Framework of the post-hoc XAI approach for IIoT intrusion detection model explainability. It illustrates the workflow process of the attack classification, XAI interpretation of prediction model outcome, and quantitative evaluation of model judgments, making informed decisions available to cybersecurity experts and stakeholders.

a subset of the most relevant features describes the predictions. Representative models are trained on the perturbed data, focusing on the selected features. The feature weights obtained from these models explain the observation, shedding light on how specific features influence the prediction. Figure 1 details the process of the post-hoc XAI.

Post-Hoc SHAP and LIME Explainability of Machine Learning-Based Attack Classification

Research has demonstrated the effectiveness of ML classifiers for IIoT intrusion detection despite their complex and difficult-to-interpret "black box" nature. With interconnected layers and complex operations, these models challenge understanding their decision-making process, potential biases, and trustworthiness. To overcome these challenges, this study proposes a post-hoc explainability approach that employs the prevalent SHAP and LIME explainability techniques in intrusion detection models, focusing on feature importance and local interpretability of the state-of-the-art predictive algorithms. Recent advancements in XAI have demonstrated significance; evaluating post-hoc approaches for ML-based IDSs in IIoT time-critical infrastructures is imperative. The WUSTL-IIoT dataset trains the classification algorithms, where features represent system behavior characteristics, and labels indicate if an instance is benign or attacked. These models classify new instances and detect potential attacks by leveraging the learned patterns and features. The trained algorithms were significant in attack detection.

Then, the post-hoc explainability framework of SHAP and LIME techniques determines influential features through feature importance and local interpretability by approximating the behavior of black box models to explain individual predictions. SHAP measures the contribution of each feature by extending the integrated gradients method, while LIME approximates complex models locally using interpretable models [5, 7, 13]. The framework involves determining an interpretable representation, drawing a perturbed sample, applying the original model, fitting an interpretable model to the data, and assessing the relevance of interpretable components.

The high point of this study is that the evaluation of the explainability framework validates a subjective analysis of the model decisions and users' trust. The confidence and decision impact ratios enabled this achievement [7] The decision impact ratio is the rate of change in judgments due to the omissions of critical features in the interpretation method. At the same time, the confidence impact ratio indicates the potential decline in reliance due to the omission of a vital feature in the interpretation approach.

Contrasting Post-Hoc XAI with Emerging Techniques

Post-hoc explainability aims to interpret model outputs without modifying the model, while causal reasoning, often considered during model development, focuses on understanding underlying causal relationships. Causal reasoning delves beyond correlations, using techniques like causal inference algorithms and counterfactual reasoning. In contrast, post-hoc explainability provides insights into a model's decision-making without altering it and applies to any existing model. However, causal reasoning requires careful consideration and additional data for establishing causal relationships. The choice depends on the target domain, analysis goals, and the desired level of understanding.

Experimentation Environment and Result Analysis

This section covers the experimental setup, models, and datasets used in the study, emphasizing the evaluation of various classifiers and the proposed post-hoc XAI architecture. The focus is on local explanations and feature relevance, assessing the effectiveness of generated explanations and providing insights into the data features crucial for model predictions.

This study conducted methodological experimentation to analyze the considered classifiers' predictions and the XAI model judgments. The classifiers employed in this study are for binary and multi-classification of IIoT attacks in the WUSTL-IIoT dataset, assessing the interpretability capacity of each visible explainability technique by quantifying their confidence and decision impact ratios.

IIoT network security has ensured using ML solutions and big data analytics. However, cyber risks vary because IIoT and traditional networks

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Algorithm	Binary		Multi-class	
	Loss (#)	Computation Time (s)	Loss (#)	Computation Time (s)
Decision Tree	0.0002	0.52	0.002	0.37
Random Forest	5.227	24.74	7.610	6.20
Deep Neural Network	0.0067	1223.66	0.0041	2189.67
Transformer	0.00097	503.40	9.2135e-04	405.63

TABLE 2. Computation time and loss performance of evaluated algorithms.

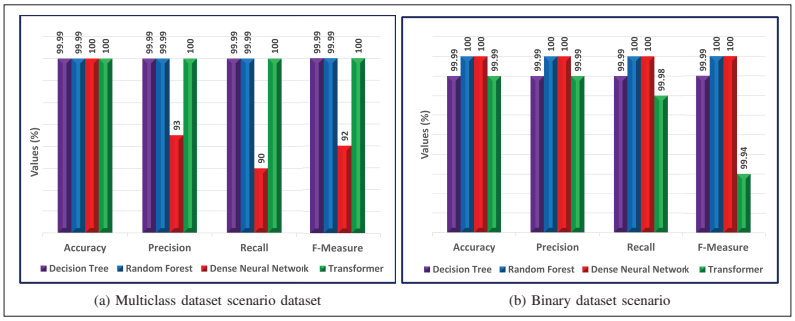


FIGURE 2. Graph of prediction performance evaluation metrics on the: a) multi-class; b) binary scenarios of the WUSTL-IIoT dataset.

have fundamentally different functions and priorities. Given the evolving nature of cybersecurity and IDS for industrial networks, studies have shown the effectiveness of deep neural networks and tree-based algorithms for attack detection in IIoT [1, 14], motivating the representative choice of the decision tree, random forest, deep neural network, and transformer models in this study.

Evaluation Dataset Selection and Preprocessing

This study leverages the WUSTL-IIoT[1] datasets comprising IIoT network data for cybersecurity research.

Data Selection and Justification: The dataset was selected based on its relevance to cyber-attacks in IIoT networks. The dataset includes various IoT attacks, such as distributed denial of service, command injection, backdoors, and reconnaissance. The dataset size is approximately 2.7 GB and covers about 53 hours of data samples. It was generated using the IIoT testbed [14], designed to closely mimic real-world industrial systems and enable the execution of authentic cyber-attacks. The model experimentation was with total data observations of 1,194,464, comprising 1,107,448 normal samples, 87,016 attack samples, and 41 data features split using the train-test-split modules in Keras and Scikit-learn in the proportion of training (60%), testing (25%), and validation (15%), respectively, for reproducibility.

Data Preprocessing: To improve model efficiency, the feature selection process during the data preprocessing phase utilized the Pearson coefficient correlation algorithm to calculate the relationships between data features, eliminating uncorrelated features. Setting a correlation threshold of 0.4 resulted in 18 relevant and correlated features with 1 depicting a positive linear correlation and a zero correlation by 0. StandardScaler enabled obfuscation of the relationship between the original feature values and the classification to prevent direct inference of classification data. The experimentation environment is Python on Google Colaboratory on an Intel(R) Core(TM) i5-8500 CPU @ 3.00GHz system with 8GB RAM running Windows 11.

[1] https://www.cse.wustl.edu/jain/iiot2/index.html

Rationale for Post-Hoc LIME and SHAP Interpretation

The use of SHAP and LIME in this study is motivated by their empirically supported interpretation, which provides a solution with significant representativeness, fidelity, and expressed influence. The approach also incorporates global and local interpretations, making it the superior choice of methods for model interpretations [5, 11, 12]. They are also the most popular algorithms for determining model explainability. LIME has a desirable additivity property, but it also has shortcomings in uniformity, missingness, and stability that SHAP fills in; hence, it is popular.

Model Performance and Evaluation

The metrics employed to evaluate the classification models and XAI as demonstrated in [10] are discussed in the following subsections. These metrics visually elucidated the predictive capabilities, showing the extent of understanding and trust in XAI for algorithms.

IDS Evaluation Metrics: The performance of evaluated classification algorithms was assessed using accuracy, f-measure, precision, and recall. The f-measure balances precision and recall to evaluate the model and performs better than accuracy, especially in heterogeneous IIoT scenarios with multi-class and sample-class imbalances. The preference for f-measure over accuracy in this study is due to the latter's lack of consideration in class distribution, which is often misleading in imbalanced datasets.

XAI Evaluation Metrics: This study analyzed the effects of LIME and SHAP interpretation methods using decision and confidence impact ratios metrics (Fig. 1) and mean votes. The mean vote provides an average score based on human evaluators' judgments of the explanations generated by the XAI techniques for different algorithms. The decision impact ratio is the overall influence of the data features on the model's decision, while the confidence impact represents the level of dependence on the prediction model.

IDS Performance Evaluation: F-measure consolidates precision and recall into a unified metric, essential in the IIoT environment to correctly identify the minor attack instances while minimizing false positives and negatives, as demonstrated by the loss performance in Table 2. Figure 2 illustrates the prediction performance of the various models' outcomes to the test dataset in IIoT attack prediction. The decision tree algorithm demonstrated consistency in significantly predicting various attack types IIoT attacks in both (a) multi-class and (b) binary scenarios with an f-measure of 99%, the least computation time, and minimal error loss. The (92%) recall of the deep neural network in (a) validates the sensitivity of the f-measure to accurately predict all attack classes in the multi-classification scenario with class imbalance. The classification algorithms of decision trees, random forests, transformers, and deep neural networks individually integrate the XAI techniques for binary and multi-class scenarios. The obtained relevant features influenced the generation of binary (malicious, benign) and multi-attack classes of backdoors, command injection, denial of service, reconnaissance, and normal traffic.

XAI Performance Evaluation: The LIME and SHAP XAI approaches enabled the assessment

Algorithm	XAI Approach	Binary		Multiclass	
		Decision Impact Ratio	Confidence Impact Ratio	Decision Impact Ratio	Confidence Impact Ratio
Decision Trees	LIME	0.8500	0.1500	0.9050	0.0950
	SHAP	0.4000	0.6000	0.1113	0.8887
	Mean Votes	0.625	0.375	0.508	0.491
	Computation Time (s)	**0.033**		**0.049**	
Random Forests	LIME	0.9066	0.0934	0.9023	0.0977
	SHAP	0.3120	0.5880	0.0078	0.7922
	Mean Votes	0.609	0.340	0.455	0.444
	Computation Time (s)	**0.076**		**0.15**	
Dense Neural Networks	LIME	0.9558	0.0442	0.9566	0.0434
	SHAP	0.1605	0.8395	0.0203	0.3600
	Mean Votes	0.559	0.442	0.488	0.2017
	Computation Time (s)	**0.5**		**1.19**	
Transformer	LIME	.9558	0.0442	0.9566	0.0434
	SHAP	0.4264	0.5736	0.4561	0.543
	Mean Votes	0.6911	0.3089	0.7064	0.2937
	Computation Time (s)	**21.62**		**7.41**	

TABLE 3. Explainability analysis of the Post-hoc LIME and SHAp interpretation approaches based on the f-measure of the evaluated state-of-the-art algorithms.

and interpretation of the model prediction scores for the normal traffic and attack types. The mean votes of the interpretation results of the LIME and SHAP were ascertained based on the data features and prediction model dependency. It is crucial to acknowledge that all experimentation was within a universal setting where judgments are by the identical network on identical data.

Table 3 shows the experimentation results of the decision and confidence impact ratio metrics of the LIME and SHAP explainability. The quantitative evaluation of LIME and SHAP interpretation shows that LIME achieves a higher decision impact over the SHAP explainer. It indicates that the decision of the LIME explainability on the model detection capability depends on the data features. Figure 3 demonstrates the significant contributory features that impacted predicting the various attacks. The confidence impact, on the other hand, represents the reliance on the model detection ability.

Similarly, the SHAP decision impact ratio shows its dependence on the data features and vice versa. The standard deviation of the decision and confidence impact ratios of the LIME and SHAP interpretation analysis gave the mean votes. If the decision impact ratio is either equal to or exceeds the confidence impact ratio, the mean vote is 1; else, it is 0. It enhances decision-making and model inference for cybersecurity experts without the necessary mathematics or AI knowledge compared to study [3]. Additionally, the mean votes and computation time in Table 3 reveal that LIME highlights specific data features significantly influencing the model's decision, providing quick results. LIME's insights are crucial for cybersecurity experts and stakeholders to identify critical factors driving the model's predictions. In

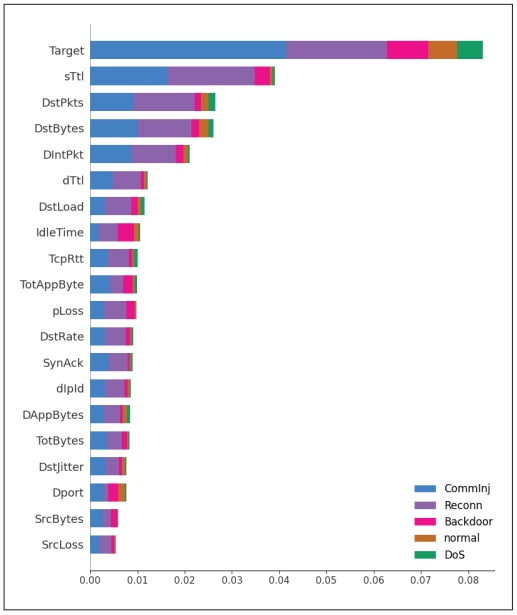


FIGURE 3. Average feature impact on the detection of the four attack classes and normal traffic.

contrast, SHAP's lower decision impact ratio suggests less reliance on its analyzed features, indicating the model's dependence on broader factors or different features. While SHAP offers a comprehensive understanding of feature contributions, its features impact the model's decision less.

Summary of Key Findings

Analyzing decision impact ratios between LIME and SHAP provides insights into varying degrees of feature importance for IIoT cybersecurity,

The standard deviation of the decision and confidence impact ratios of the LIME and SHAP interpretation analysis gave the mean votes. If the decision impact ratio is either equal to or exceeds the confidence impact ratio, the mean vote is 1; else, it is 0.

aiding in efficient computation and model interpretation. This understanding supports real-time decision-making in attack mitigation, fostering consensus among IIoT users and experts and promoting trust [10]. LIME emphasizes strong data feature influence but lower confidence, aiding in understanding critical factors and decision-making. In contrast, SHAP offers a holistic view with lower reliance on specific features but higher overall confidence, highlighting each feature's contribution. Choosing between LIME and SHAP should align with specific needs and goals, considering focus, confidence, interpretability, model behavior analysis, and complexity for informed decision-making.

Conclusion

This study integrates XAI techniques, specifically LIME and SHAP, with ML-based intrusion detection algorithms for modeling IIoT network attack detection to enhance trust and transparency. Evaluation through decision and confidence impact ratios reveals that LIME and SHAP provide valuable insights into IIoT attack detection, supporting informed cybersecurity decisions. Mean votes emphasize the significance of these XAI methods in interpreting IIoT attack predictions. Prioritizing threat analysis based on influential features and assessing system reliability through confidence levels, LIME and SHAP contribute to cognizant decision-making in IIoT cybersecurity. The study offers actionable insights for experts, enabling real-time, informed decisions and enhancing the precision of cybersecurity measures. Overcoming the inherent complexity of IIoT systems is a critical challenge in XAI due to the diversity of data sources and the dynamic nature of cyber threats. Developing XAI techniques tailored to the unique characteristics of IIoT cybersecurity, considering factors such as the diversity of data and the specificity of the industrial context, is imperative. Future directions involve enhancing model adaptability and incorporating user-friendly interfaces for optimal XAI integration in decision-making.

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